PPO Summary: From Policy Gradient Theorem to State-of-the-Art GAE Implementation

This document provides a concise summary of the 8-step mathematical journey from the foundational policy gradient theorem to the sophisticated Generalized Advantage Estimation (GAE) used in modern PPO implementations.

Step 1: Policy Gradient Foundation

Core Objective: Maximize expected return $J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)]$

Key Assumptions:

- MDP Framework: States S, actions A, transitions $P(s'|s,a)$, rewards $R(s,a)$, discount γ
- Differentiability: Policy $\pi_\theta(a|s)$ differentiable w.r.t. θ
- Stationarity: Time-invariant dynamics and policy (momentarily during gradient computation)

Policy Gradient Theorem:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) R_t\right]$$

Key Equations:

- Return: $R_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$ (causality principle)
- Likelihood ratio trick: $\nabla_\theta \mathbb{E}_{x \sim p_\theta}[f(x)] = \mathbb{E}_{x \sim p_\theta}[f(x) \nabla_\theta \log p_\theta(x)]$

Rationale: Uses likelihood ratio trick and causality principle to derive unbiased gradient estimator. Log probability provides numerical stability, natural gradient interpretation, and mathematical elegance.

Step 2: The High Variance Problem and Baseline Subtraction

Problem: Basic policy gradient suffers from extremely high variance, making learning unstable.

Key Concepts:

- **Variance**: Gradient estimates fluctuate around expected value across trajectory samples
- **Bias**: Systematic error between estimator expectation and true gradient

Solution - Baseline Subtraction Theorem:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)(R_t - b(s_t)) \right]$$

Key Insight: $\mathbb{E}_{a \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a|s) b(s)] = 0$

Optimal Baseline: $b^*(s) \approx V^\pi(s) = \mathbb{E}_{a \sim \pi_\theta}[Q^\pi(s,a)]$

Rationale: Any state-dependent baseline can be subtracted without introducing bias, with state value function being the variance-minimizing choice.

Step 3: Emergence of Advantage Functions

Mathematical Derivation: Using $V^\pi(s)$ as baseline leads to advantage functions:

$$A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$$

Advantage-Based Policy Gradient:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) A^\pi(s_t, a_t) \right]$$

Critical Property: $\mathbb{E}_{a \sim \pi_\theta}[A^\pi(s,a)] = 0$ (zero-mean)

Mathematical Proof:

$$\mathbb{E}_{a \sim \pi_\theta}[A^\pi(s,a)] = \mathbb{E}_{a \sim \pi_\theta}[Q^\pi(s,a)] - V^\pi(s) = V^\pi(s) - V^\pi(s) = 0$$

Rationale: Advantages provide relative action quality information without introducing bias, ensuring unbiased policy gradient updates while reducing variance.

Step 4: Temporal Difference (TD) Estimation

Problem: Direct computation of $Q^\pi(s,a)$ and $V^\pi(s)$ is intractable.

TD Solution: Use Bellman equation to estimate advantages:

$$\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$$

Key Assumptions:

- Bellman Validity: $V^\pi(s_t) = \mathbb{E}_\pi[r_t + \gamma V^\pi(s_{t+1})|s_t]$
- Markov Property: Future depends only on current state
- Value Function Accuracy: $V(s) \approx V^\pi(s)$

Bias-Variance Tradeoff:

- **Monte Carlo Direction**: Low bias, high variance (using actual rewards)
- **TD Direction**: High bias, low variance (using bootstrapped estimates)

Advantage Approximation: When $V(s) = V^\pi(s)$, then $\delta_t^V = A^\pi(s_t, a_t)$

Rationale: TD errors provide computationally tractable advantage estimates, introducing the fundamental bias-variance tradeoff in advantage estimation.

Step 5: The Bias-Variance Tradeoff in Advantage Estimation

Core Dilemma:

- **TD Error ($\lambda = 0$)**: $\hat{A}_t = \delta_t \rightarrow$ High bias, Low variance
- **Monte Carlo ($\lambda = 1$)**: $\hat{A}_t = \sum_{k=0}^{\infty} \gamma^k \delta_{t+k} \rightarrow$ Low bias, High variance

High Variance Sources:

1. Stochastic rewards and environment transitions
2. Function approximation errors
3. Finite sampling

Bias Sources:

1. Value function approximation: $V_\phi(s) \neq V^\pi(s)$
2. Bootstrapping with estimated values
3. Finite horizon truncation

Rationale: Understanding the sources of bias and variance enables principled design of advantage estimators that balance these competing objectives.

Step 6: Generalized Advantage Estimation (GAE) - The Optimal Solution

Mathematical Foundation:

$$\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V$$

Recursive Implementation:

$$\hat{A}_t^{GAE} = \delta_t + \gamma\lambda \hat{A}_{t+1}^{GAE}$$

Bias-Variance Control Parameter λ:

- $\lambda = 0$: Pure TD → High bias, low variance
- $\lambda = 1$: Pure MC → Low bias, high variance
- $\lambda \in (0, 1)$: Optimal tradeoff → Exponentially weighted combination

Key Assumptions:

1. Exponential decay: Future information becomes less reliable exponentially
2. Geometric convergence: $|\gamma\lambda| < 1$ ensures series convergence
3. Finite horizon: Practical truncation at episode boundaries

Rationale: GAE provides principled interpolation between bias and variance, allowing practitioners to tune the λ parameter for optimal performance in specific domains.

Step 7: State-of-the-Art Implementation in PPO

Practical GAE Algorithm:

```
import jax.numpy as jnp

def gae_advantages(rewards, terminal_masks, values, discount, gae_param):
    advantages = []
    gae = 0.0
    for t in reversed(range(len(rewards))):
        delta = rewards[t] + discount * values[t + 1] * terminal_masks[t] - values[t]
        gae = delta + discount * gae_param * terminal_masks[t] * gae
        advantages.append(gae)
    return jnp.array(advantages[::-1])
```

Why Reversed Loop is Essential:

- **Mathematical Dependency**: $\hat{A}_t^{GAE} = \delta_t + \gamma\lambda\hat{A}_{t+1}^{GAE}$ creates backward dependency chain
- **Boundary Condition**: $\hat{A}_T^{GAE} = \delta_T$ (terminal condition)
- **Computational Efficiency**: $O(T)$ linear time complexity
- **Terminal State Handling**: Prevents advantage bleeding across episodes

Implementation Assumptions:

- Neural network approximation: $V_\phi(s) \approx V^\pi(s)$
- Backward processing for efficient computation
- Terminal state handling: $V(s_{\text{terminal}}) = 0$
- Batch processing across parallel environments

Hyperparameter Sensitivity:

- $\gamma \in [0.95, 0.99]$: Discount factor
- $\lambda \in [0.9, 0.99]$: GAE parameter

Rationale: Efficient implementation that maintains mathematical correctness while enabling practical deployment in modern RL systems.

Step 8: Integration with PPO Loss Function

Final Policy Gradient with GAE:

$$\nabla_\theta J(\theta) \approx \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \hat{A}_t^{GAE} \right]$$

PPO Clipped Objective:

$$L^{CLIP}(\theta) = \mathbb{E}_t[\min(r_t(\theta)\hat{A}_t^{GAE}, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t^{GAE})]$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the importance sampling ratio.

Key Components:

1. **Importance Sampling Ratio**: Enables off-policy updates and sample reuse
   - Corrects for policy mismatch between data collection and updates
   - Mathematical foundation:

$$\mathbb{E}_{\tau \sim \pi_\theta}[f(\tau)] = \mathbb{E}_{\tau \sim \pi_{\theta_{old}}} \left[\frac{\pi_\theta(\tau)}{\pi_{\theta_{old}}(\tau)} f(\tau) \right]$$

   PPO's Hybrid On-Policy/Off-Policy Approach:
   The use of importance sampling in PPO represents a **perfect example** of the on-policy/off-policy discussion in reinforcement learning, creating a sophisticated hybrid approach:
   Traditional Distinction:
   - **On-Policy Methods**: Learn about the policy being used to make decisions (e.g., REINFORCE, SARSA)
     - ✅ Theoretical guarantees
     - ✅ Stable training
     - ❌ Sample inefficient
   - **Off-Policy Methods**: Can learn from data generated by different policies (e.g., Q-learning, DQN)
     - ✅ Sample efficient
     - ✅ Can learn from any data
     - ❌ Can be unstable
     - ❌ Distribution mismatch issues

   PPO's Innovation: PPO bridges this gap through its two-phase process:
   - **Experience Collection**: Use policy $\pi_{\theta_{\text{old}}}$ (on-policy data collection)

- **Multiple Updates**: Reuse same data for several gradient steps (off-policy learning with importance sampling correction)

The Fundamental Tradeoff PPO Solves:

```
Traditional On-Policy: π_θ_old → collect data → one update → π_θ_new → discard data → repe
PPO Approach: π_θ_old → collect data → multiple updates → π_θ_new (reuse same data!)
```

Why This Matters:

- **Sample Efficiency**: Can reuse data multiple times (like off-policy methods)
- **Stability**: Clipping prevents destructive updates (like on-policy methods)
- **Practical Performance**: Works well across diverse domains
- **Theoretical Soundness**: Maintains convergence guarantees

This hybrid approach demonstrates that the on-policy/off-policy distinction isn't binary but rather a spectrum, with PPO finding an optimal balance between sample efficiency and training stability.

2. **Clipping Mechanism**: Prevents destructively large policy updates

$$\text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) = \begin{cases} 1-\epsilon & \text{if } r_t(\theta) < 1-\epsilon \\ r_t(\theta) & \text{if } 1-\epsilon \leq r_t(\theta) \leq 1+\epsilon \\ 1+\epsilon & \text{if } r_t(\theta) > 1+\epsilon \end{cases}$$

3. **Complete PPO Loss:**

$$L_{PPO}(\theta) = L^{CLIP}(\theta) - c_1 L^{VF}(\theta) + c_2 S[\pi_\theta]$$

Theoretical Properties:

- Conservative updates through clipping
- Monotonic improvement guarantees
- Sample efficiency via experience reuse
- Training stability through bounded policy changes

Rationale: Integration of GAE with PPO's clipped objective creates a theoretically sound and practically effective algorithm that balances sample efficiency, training stability, and performance.

Summary of Key Benefits

1. **Variance Reduction**: GAE smooths noisy advantage estimates
2. **Bias Control**: λ parameter enables fine-tuning of bias-variance tradeoff
3. **Sample Efficiency**: Better gradient estimates accelerate learning
4. **Stability**: Reduced gradient noise and clipping improve training stability
5. **Credit Assignment**: Multi-horizon consideration improves temporal credit assignment

Theoretical Guarantees

- **Unbiasedness**: When $V(s) = V^\pi(s)$, GAE provides unbiased advantage estimates
- **Consistency**: As $\lambda \to 1$, GAE approaches Monte Carlo estimates
- **Convergence**: Maintains policy gradient theorem guarantees while improving practical performance

This progression represents a principled evolution driven by the fundamental bias-variance tradeoff, culminating in state-of-the-art advantage estimation used in modern PPO implementations.

Practical Tensor Structures for PPO Optimization

This section ties the 8-step conceptual journey to the concrete tensors that flow into the PPO loss in this NNX-based implementation.

- Collection across parallel environments and time:
  - We run N parallel simulators for T steps to collect experience.
  - Raw stacks (before flattening):
    - states: shape (T, N, 84, 84, 4)
    - actions: shape (T, N) (discrete action indices)
    - rewards: shape (T, N)
    - values: shape (T+1, N) (bootstrap one extra value for GAE)
    - old_log_probs: shape (T, N) (log π_{θ_old}(a_t|s_t))
    - dones: shape (T, N) (1.0 at terminal, else 0.0)
- Per-environment GAE and returns (Step 6–7):
  - For each environment i ∈ {1,…,N}, form terminal masks m_t = 1 − done_t.
  - Call GAE with vectors of length T:
    - rewards[:, i] → (T,)
    - terminal_masks → (T,)
    - values[:, i] → (T+1,)
  - Compute for each env i:
    - advantages[:, i]: shape (T,)
    - returns[:, i] = advantages[:, i] + values[:-1, i]: shape (T,)
- Flatten for optimization (Step 8):
  - Define B = T × N (the total number of time-steps across all envs).
  - The training trajectories passed to the optimizer are flattened to:
    - states: (B, 84, 84, 4)
    - actions: (B,) int32
    - old_log_probs: (B,) float32
    - returns: (B,) float32
    - advantages: (B,) float32 (normalized inside the loss)
- Minibatching for the loss:
  - During an epoch, the flattened arrays are reshaped to (iterations, batch_size, …) and iterated.

- The loss function consumes one minibatch tuple:
    - states: (batch_size, 84, 84, 4)
    - actions: (batch_size,)
    - old_log_probs: (batch_size,)
    - returns: (batch_size,)
    - advantages: (batch_size,)
  - Model outputs for discrete actions:
    - log_probs: (batch_size, num_actions)
    - values: (batch_size, 1) → squeezed to (batch_size,)
- What the PPO loss computes per minibatch (high level):
  - Gather log_probs_act_taken = log_probs[j, actions[j]] per sample.
  - Ratio r_t = exp(log_probs_act_taken – old_log_probs).
  - Normalize advantages: (A – mean(A)) / (std(A) + 1e–8).
  - Policy loss = –mean(min(r_t·A, clip(r_t, 1–ε, 1+ε)·A)).
  - Value loss = mean((returns – values)^2).
  - Entropy bonus = mean categorical entropy of the policy.

Concrete examples

1. Typical Atari-like setup (discrete actions)

- Suppose T = 128 rollout steps and N = 8 environments, so B = 1024.
- num_actions = 4 (example).
- After flattening:
  - states.shape = (1024, 84, 84, 4), dtype=float32
  - actions.shape = (1024,), dtype=int32, e.g. [1, 0, 3, 2, 1, …]
  - old_log_probs.shape = (1024,), e.g. [–0.69, –1.20, –0.92, …]
  - returns.shape = (1024,), e.g. [0.43, 0.51, –0.12, …]
  - advantages.shape = (1024,), normalized to mean≈0 and std≈1
- A single minibatch with batch_size = 256 would have the same per-field shapes with 256 instead of 1024.

2. Tiny end-to-end toy snapshot (T = 2, N = 3 ⇒ B = 6)

- Flattened tensors passed into one loss evaluation (batch_size = 6):
  - states: (6, 84, 84, 4)
  - actions: [2, 0, 1, 3, 1, 0]
  - old_log_probs: [–1.20, –0.69, –1.10, –1.39, –0.75, –1.25]
  - returns: [0.97, 0.12, –0.05, 0.22, 0.08, –0.11]
  - advantages (before normalization): [0.50, –0.10, 0.20, –0.30, 1.00, –1.30]
  - advantages (after normalization inside loss): mean≈0, std≈1
- The model produces:
  - log_probs: (6, num_actions)
  - values: (6, 1) → (6,)
- The loss then computes r_t, applies clipping with ε, value MSE to returns, and adds an entropy bonus.

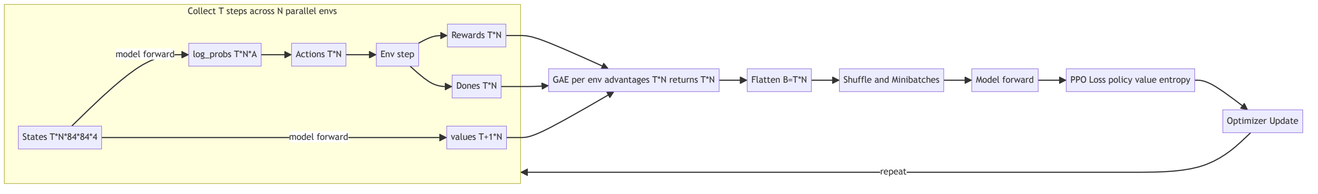
Notes and edge cases

- Terminal handling: masks prevent advantages from leaking across episode boundaries.
- Dtypes: actions are integer indices; all other scalars are float32.
- Continuous action variants would use actions shaped (B, action_dim) and log-prob scalars per sample, but this example uses discrete actions (categorical policy).

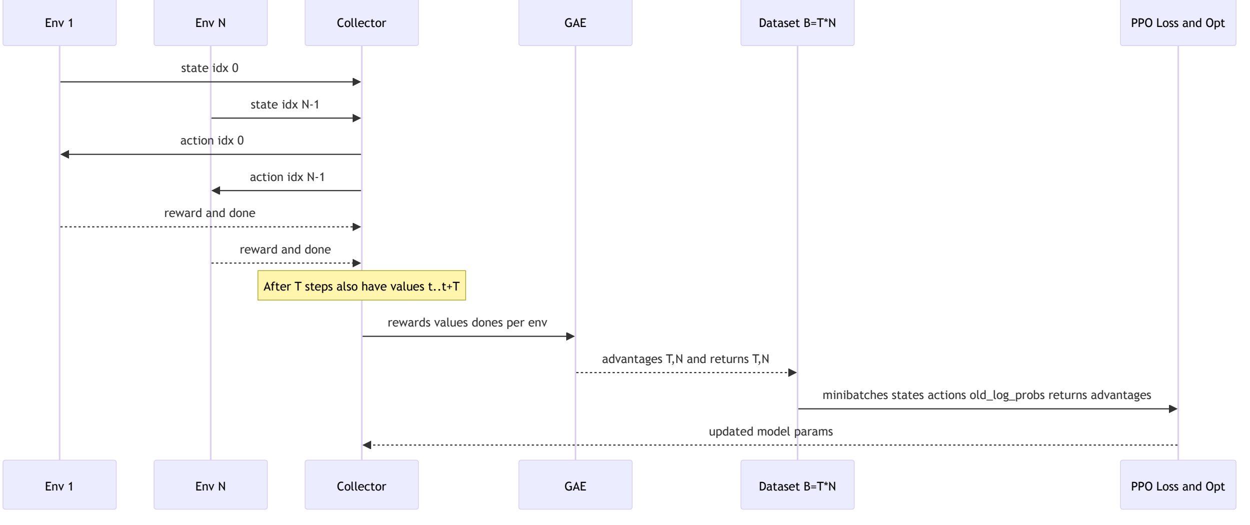
Visualizing the PPO Data Process

A concise way to understand how tensors move through PPO is to visualize the pipeline from rollout collection to the optimized update.

Flowchart overview



End-to-end interaction (sequence)



What to plot while debugging or explaining the flow

- Advantages histogram: expect mean≈0 after normalization, track std.
- Policy ratio r_t histogram and clip fraction: how often clipping is active.
- Value loss and returns vs. predicted values scatter: calibration of critic.
- Entropy and action distribution: policy exploration over time.
- Episode length/reward curves: aligns with data segmentation and masks.

Tip: These can be logged with TensorBoard; this repo already writes scalar game_score. Adding histograms for advantages and ratios in the training loop can make these plots immediately available.